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CSEN 1144: Generative and Agentic AI, Spring Term 2026
Assignment 2: Agentic Wordle
Due May 12, 2026

1. Assignment Description

In Lecture 9, you studied three reasoning strategies: Chain-of-Thought, Self-Reflection, and Self-Consistency. In this assignment you will test all three strategies empirically by building agents in LangGraph that play the popular Wordle game and observing whether reasoning strategies actually make a difference.

The objective of the agent playing Wordle is to guess a hidden 5-lettered word in a maximum of six attempts. In every attempt, the agent makes a guess by entering any 5-lettered word. After each guess, a color-coded feedback is returned for every letter with green meaning correct letter and correct position, yellow meaning correct letter but wrong position, and gray meaning incorrect letter. The agent must use the feedback from each guess to eliminate impossible words and converge on the answer.

2. Required Agents

You are required to build four agents in **LangGraph**. Each uses the same state schema, the same game environment, and the same model. The only difference is the reasoning strategy applied inside the agent node.

Agent 1: Baseline. The agent receives the feedback history and generates a guess directly. No explicit reasoning instructions beyond the game rules.

Agent 2: Chain-of-Thought. Before guessing, the agent explicitly reasons through what it knows (which letters are confirmed, which are eliminated, which are misplaced) and then generates a guess consistent with those constraints.

Agent 3: Self-Reflection. After generating a candidate guess, the agent critiques it before submitting. The critique checks whether the guess violates any known constraints. If it does, the agent revises. Only then is the guess submitted to the game.

Agent 4: Self-Consistency. The agent generates three independent candidate guesses and selects the most common one. If all three are different, it selects the one that satisfies the most constraints from the feedback history.

3. Experimental Design

After implementing your agents, run all four agents on the same five hidden words: *crane, light, proud, storm, blaze.* Every agent must attempt every word. For each agent and word, record whether the agent solved the puzzle, the number of guesses used, and the sequence of guesses made. Print a summary table and the win percentage for each agent. Note that self-Reflection and self-Consistency are significantly slower than the baseline due to multiple model calls per guess. A good strategy is to develop and test on one word first before running the full experiment.

4. Deliverables

You are required to submit an `ipynb` notebook with the implementation of the agents in LangGraph and the experiment as described. Include a markdown at the end of the notebook answering the following questions.

a) Which reasoning strategy performed best?

b) What surprised you?

c) What is the biggest limitation of your implementation?

d) How would you fix it?

e) Reflection: Do reasoning strategies help for this type of game?

5. Groups

This is an individual assignment.